

ATHRAV SERUWAM

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SUMMARY OF TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, HTML/CSS

Frameworks & Tools: React, Next.js, Node.js, Tailwind CSS, Streamlit, RESTful APIs, Git, GitHub

APIs & Platforms: OpenAI, Stripe, Pinecone, Vercel

EXPERIENCE

Software Engineering Intern

Jan 2026 – Present

Quotograph

Kitchener, ON

- Developed zero-shot computer vision workflows using VLMs to detect architectural objects (e.g., fixtures, rooms) in floorplans without labeled training data
- Designed backend APIs in FastAPI to orchestrate model inference, post-processing, and measurement outputs
- Integrated AI pipelines into a Next.js product frontend, delivering production-ready features end-to-end

Firmware Developer

Sept 2025 – Present

University of Waterloo Formula Electric

Waterloo, ON

- Develop embedded **C/C++ firmware** for ECUs managing vehicle communication via **CAN/LIN networks** and sensor interfaces
- Implement **signal simulation**, **HIL testing**, and **fault diagnostics** to validate control board performance and ensure system reliability, reducing per-board validation time by **35%**
- Collaborate with electrical and controls teams to debug hardware, optimize signal timing, and enhance real-time responsiveness

Software Engineering Fellow

July 2024 – Sept 2024

Headstarter AI

Remote

- Engineered **AI-driven** web applications, leveraging **Next.js**, **OpenAI API**, **Stripe**, and **Pinecone** to integrate full-stack and applied ML workflows
- Designed an **AI Customer Support Chatbot** powered by GPT-4, achieving a **30% faster response rate** in testing and recognized for UI/UX innovation
- Implemented **state management and API integration** for dynamic data handling, improving reliability and scalability
- Utilized **Git** and **GitHub** for version control, collaborating through pull requests and code reviews

PROJECTS

AI Flashcard Generator

- Built a **Python** app with **Streamlit** and **LLMs (Ollama/OpenAI)** to automatically generate study flashcards from lecture notes, PowerPoints, and PDFs
- Applied **chunking**, **NLP techniques**, and **prompt engineering** to create clear, context-aware Q/A pairs
- Designed an intuitive, user-friendly interface with integrated study mode and export to Anki/CSV
- Integrated PDF and PowerPoint parsing modules to extract and process content for flashcard generation

Daily Schedule Optimizer

- Built a responsive time-blocking web app using **Next.js**, **Tailwind CSS**, and **Framer Motion**, generating optimized daily schedules from user-defined tasks
- Implemented a **priority-based scheduling algorithm** with automatic break insertion, overflow handling, and CSV/ICS export for seamless calendar integration
- Deployed on **Vercel**, delivering a smooth, interactive user experience across devices and streamlining daily planning efficiency

LEADERSHIP & EXTRACURRICULARS

DECA Club

Sept 2023 – June 2025

Vice President

Garth Webb Secondary School

- Mentored **50+ students** in business research, report writing, and presentation skills, helping 7 students qualify for **DECA Internationals (ICDC)**
- Coordinated 3+ fundraising campaigns, **raising over \$2,000** to support competitive events
- Guided peers in **business problem-solving** and effective communication under competition deadlines

EDUCATION

University of Waterloo

Sept 2025 – Apr 2030 (Expected)

Candidate for BAsC, Mechatronics Engineering

Waterloo, ON, Canada